

TADDLEBOX — UI/UX Document

Version 1.0 · React Native (Expo) · Platform: iOS & Android

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1. Design Philosophy

TADDLEBOX targets college students aged 18–25. The design reflects the energy, ambition, and social nature of campus life.

Core Design Principles

- 1. Gamified & Rewarding** Every meaningful action is visually acknowledged. XP flies to counters, cards bounce, streaks glow. The UI reinforces the sense of progress and reward.
 - 2. Dark-First, Vibrant** The default dark theme uses a deep space aesthetic (#070714 base) with electric purples, cyans, and golds as accent colors. This creates high contrast and visual excitement appropriate for gaming and social contexts.
 - 3. Layered Depth** Three surface levels (base → card → elevated) create visual hierarchy. Cards float above the background, modals float above cards. Subtle glass effects add sophistication.
 - 4. Touch-Friendly & Dense** Tap targets are minimum 40×40px. The layout is information-dense but never cluttered — spacing scales are used consistently to breathe.
 - 5. Motion as Feedback** Animations are purposeful: they confirm actions (claim → particle fly), guide attention (countdown scale), or ease transitions (modal slide). Nothing animates without reason.
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2. Design System — Colors

2.1 Dark Theme (Default)

Background Layers:

bg.base	#070714	– Root screen background
bg.surface	#0D0D1F	– Modal / sheet backgrounds
bg.card	#0F0F24	– Card backgrounds
bg.elevated	#151530	– Elevated / chip backgrounds

Brand Colors:

primary	#7C3AED	– CTA buttons, active states, XP accents
primaryLight	#A78BFA	– Text highlights, secondary accents
primaryDark	#4C1D95	– Gradient bottom / hover
cyan	#06B6D4	– Secondary brand color, progress fills
cyanLight	#67E8F9	– Light cyan highlights
cyanDark	#0891B2	– Gradient pairing with primary

XP / Reward Colors:

xpGold	#FBBF24	– XP amounts, streak counts, rewards
xpOrange	#F59E0B	– XP gradient pair

Semantic Colors:

success	#10B981	– Win states, positive amounts, verified
danger	#EF4444	– Loss states, negative amounts, errors
warning	#F59E0B	– Caution states
pink	#EC4899	– Like actions, love/heart interactions

Text Colors:

text.primary	#F1F0FF	– Main readable text
text.secondary	#A09FBF	– Supporting text, timestamps
text.muted	#5A5880	– Placeholders, disabled, labels

Border Colors:

border	rgba(255,255,255,0.06)	– Subtle card outlines
borderHover	rgba(255,255,255,0.12)	– Interactive border on focus

Glass Effects:

glass	rgba(255,255,255,0.04)	– Frosted glass backgrounds
glassBorder	rgba(255,255,255,0.08)	– Glass border

2.2 Light Theme

Background Layers:

bg.base	#F8F7FF	– Root screen background (near white, purple tint)
bg.surface	FFFFFF	– Modal / sheet backgrounds
bg.card	#F1F0FF	– Card backgrounds
bg.elevated	#E8E7F8	– Elevated / chip backgrounds

Brand Colors (same hues, adjusted for light):

primary	#7C3AED	– Unchanged (sufficient contrast on light)
primaryLight	#6D28D9	– Darker for light backgrounds
cyan	#0891B2	– Darker for light backgrounds

Text Colors (inverted):

text.primary	#1A1740	– Dark text on light backgrounds
text.secondary	#4B4870	– Medium dark
text.muted	#8B88B0	– Lighter muted

Border Colors:

border `rgba(0,0,0,0.08)` – Subtle dark outlines
borderHover `rgba(0,0,0,0.15)` – Active border

2.3 Gradient Patterns

Gradients are used extensively throughout the app to add energy and visual hierarchy:

Usage	Gradient
Primary CTA button	<code>#7C3AED</code> → <code>#0891B2</code> (left to right)
XP gold elements	<code>#FBBF24</code> → <code>#F59E0B</code>
Hero backgrounds	<code>rgba(124,58,237,0.28)</code> → transparent (top to bottom)
Community banners	Category-specific dark gradients (e.g. <code>#2a0a5e</code> → <code>#0a1f5e</code> for Tech)
Profile avatar	primary → cyanDark
Level badge	xpGold → xpOrange
Win result card	<code>rgba(16,185,129,0.15)</code> → <code>rgba(6,182,212,0.08)</code>
Loss result card	<code>rgba(239,68,68,0.15)</code> → <code>rgba(180,53,53,0.05)</code>

3. Design System — Typography

Font family: System default (San Francisco on iOS, Roboto on Android)

fontSizes:

xs 10px – Labels, badges, meta
sm 13px – Supporting text, chips, timestamps
md 15px – Body text, card content
lg 17px – Subheadings, stat values
xl 20px – Section headers
xxl 24px – Screen titles
display 42px – Hero numbers, splash text

Weights used:

- 400 — Body text
- 600 — Supporting headings, labels
- 700 — Section titles, emphasis
- 800 — Stats, values, names
- 900 — Logo ("TADDLEBOX"), display text

Text transformations:

- uppercase with `letterSpacing: 0.5` for section labels
- Numeric displays use `toLocaleString()` for comma-formatted large numbers

4. Design System — Spacing & Radii

Spacing Scale

spacing:	
xs	4px
sm	8px
md	12px
lg	16px
xl	20px
xxl	24px
xxxl	32px

Consistent usage:

- Screen horizontal padding: spacing.lg (16px) or spacing.xl (20px)
- Card internal padding: spacing.md (12px)
- Stack gap between elements: spacing.sm (8px)
- Section margin bottom: spacing.md to spacing.lg

Border Radii

radii:		
sm	8px	– Chips, small badges
md	12px	– Cards, inputs, buttons
lg	16px	– Modal sheets, info cards
xl	24px	– Bottom sheets (top corners)
full	9999px	– Pills, avatar circles, dots

5. Component Library

5.1 Button

Variants & Gradients:

Variant	Gradient / Color	Usage
primary	primary → cyanDark	Main CTAs
cyan	cyan → cyanDark	Secondary actions
xp	xpGold → xpOrange	XP-related actions
ghost	transparent + border	Tertiary actions
danger	#DC2626 → #B91C1C	Destructive actions
success	#059669 → #047857	Confirmation

Sizes:

- sm : padding 8×16, font 12px
- md : padding 12×20, font 14px (default)
- lg : padding 14×28, font 16px

States: loading (spinner), disabled (opacity 0.5), fullWidth

5.2 Input

- Background: `bg.elevated`
- Border: `1px border (resting)`, `primaryLight (focused)`
- Left icon: `Ionicons icon in text.muted`
- Error: red border + red error text below
- Password: right eye icon toggles visibility
- Label: above input, `text.secondary`, `fontSizes.sm`

5.3 PostCard

Visual structure (top to bottom):

[Avatar]	Name · @handle · timestamp
	[Community pill] [XP pill]
Post content text (1-3 lines)	
[Media banner if present] (emoji image or actual photo, 1:1/16:9)	
42	8
Share	Save

- Background: `bg.card`
- Border: `1px border (bottom only, between cards)`
- Community pill: colored background chip (`rgba(color, 0.15)`)
- XP pill: gold background `rgba(251,191,36,0.15)`
- Action icons: `text.muted (resting)`, `pink (liked)`, `primary (saved)`

5.4 Mini Card (Home Header)

7 Days
Streak)

- Flex 1 (equal width, side by side)
- Background: `rgba(color, 0.08)`, border: `rgba(color, 0.22)`
- Streak card: amber tint | XP card: purple tint
- Spring animation on XP card when reward is claimed

5.5 CustomTabBar

			[@FAB]			
Home	Comm	Events		Games	Wallet	Profile

- Background: `bg.surface` with top border

- Active tab: filled icon + `primaryLight` color + underline dot
- Inactive tab: outline icon + `text.muted`
- FAB: 52×52 gradient circle, floating above tab bar, opens `CreatePostModal`
- Safe area padding respected

5.6 XP Progress Bar

[LVL 12]  [NEXT]
Pioneer 9,240 / 12,000 XP

- Level circle: gradient (gold → orange), white level number
- Progress bar: gradient fill (purple → cyan) with glow dot at head
- Background track: `rgba(255, 255, 255, 0.07)`
- Rank text: `text.secondary`, centered

5.7 Side Drawer

- Slide from left, 80% screen width
- Backdrop: `rgba(0, 0, 0, 0.55)` semi-transparent
- Header: user gradient avatar + name + handle
- Menu items: icon + label + optional badge
- Smooth 280ms transition (delayed to prevent animation flicker)

5.8 StoryRow

- Story avatar: 60px circle
- Ring colors:
 - Own story: dashed grey outline
 - Unseen: `LinearGradient` ring (purple → pink)
 - Seen: solid grey ring
- Username: 10px text below avatar, centered, 1 line

5.9 SpotlightCarousel

- Card width: ~85% screen width
- Snap to center
- Gradient card background per item
- Dot indicators below: `text.muted` (inactive), `primaryLight` (active)
- Horizontal padding so adjacent cards peek

5.10 DailyRewardCard

- Full-width card in feed, between spotlight and posts
- Two states: unclaimed (amber tint) → claimed (green tint) → gone
- Progress track: 70% → 100% fill on claim
- Claim button: gold pill, `#1A0A00` text (dark on gold)

6. Screen-by-Screen UX

6.1 Splash Screen

Element	Behavior
Logo	Fades in and scales up with bounce

Tagline	Fades in after logo
Glow ring	Slow pulsing opacity
Transition	Auto after 2.5s → Onboarding (fade)

6.2 Onboarding Carousel

Element	Behavior
Slides	Snap scrolling, paginated
Emoji	Large centered illustration per slide
Progress dots	Animated width change on active dot
Next button	Advances or goes to Welcome on last slide
Skip	Top-right shortcut to Welcome

6.3 Home Screen

Information Hierarchy:

1. Header (status + navigation)
2. Mini cards (quick stats — streak + XP)
3. Stories (ephemeral social content)
4. Spotlight (featured/promoted content)
5. Trending chips (content filter)
6. Daily reward (time-sensitive gamification)
7. Post feed (core social content)

Scroll behavior: Full page scrolls. Header and mini-cards are pinned above the ScrollView. Stories and everything below scroll.

6.4 Game Play Modal

Phase UX:

Lobby → Countdown → Playing → Result
 (choose mode + start game) (3-2-1-GO!) (live board) (win/loss)

- Lobby: bottom sheet style inside a fullscreen modal
- Countdown: centered, full black background, pulsing number scale
- Playing: header with score, scrollable game board area
- Result: card springs in from scale 0 → 1 with opacity

6.5 Wallet Screen

Visual hierarchy:

1. Balance cards (most prominent — full-width, gradient)
2. Quick actions (earn more grid)
3. Transaction filter chips
4. Transaction list

Color coding in transactions:

- XP transactions: gold amount text
- Positive cash: green
- Negative cash (spend/withdraw): red

6.6 Community Detail

Sticky header pattern:

- Banner image is part of scroll content
- Info card overlaps banner by 20px (negative marginTop)
- Filter tabs become sticky (stickyHeaderIndices) when scrolled past

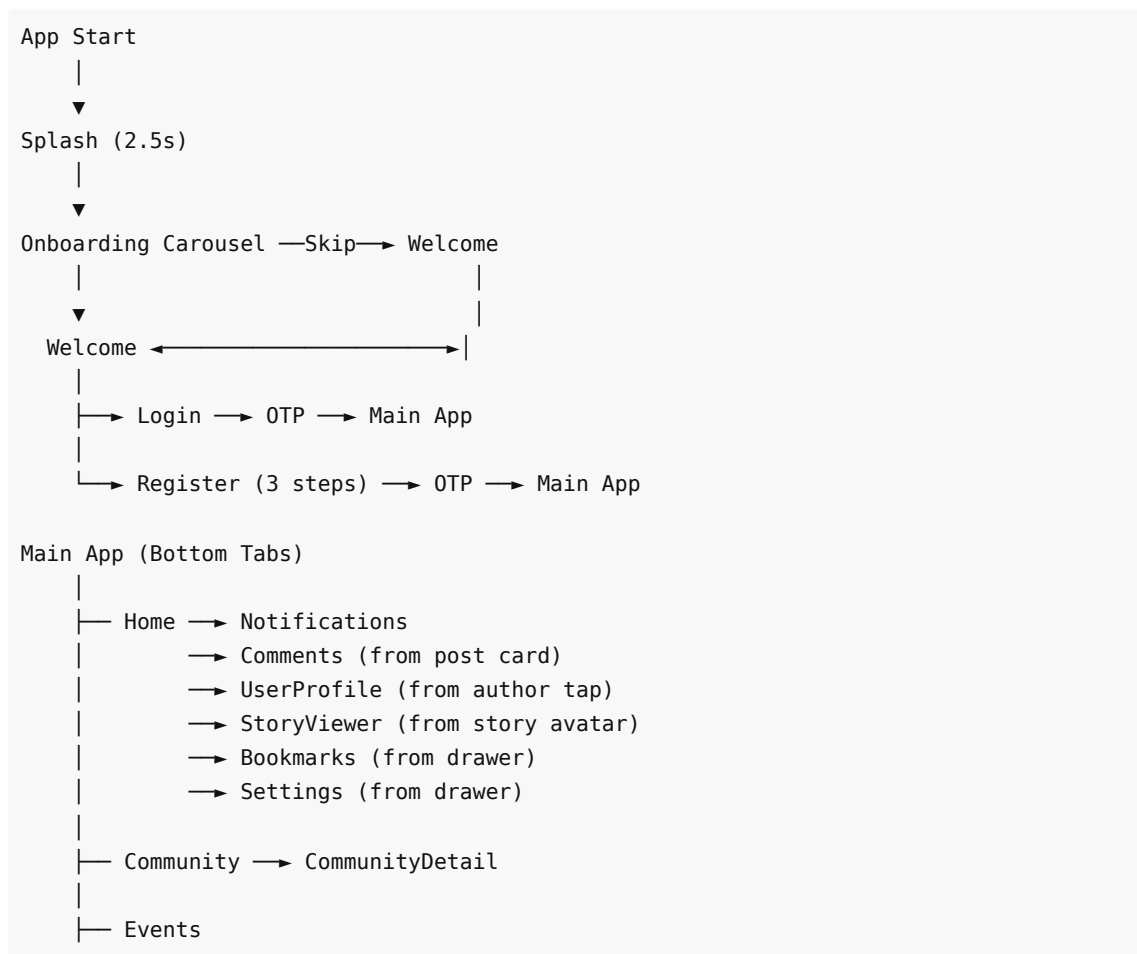
6.7 Settings Screen

Clean list-based layout with section headers:

- Each setting row: icon + label + (Toggle switch | Chevron | Value label)
- Destructive actions (logout, delete account) use `danger` color and appear at bottom
- Theme toggle has instant visual effect

7. Navigation & Flow

7.1 User Flow Diagram



- └ Games (fullscreen GameplayModal within)
- └ Wallet (modals within: Withdraw, LinkUPI, Convert, History, Settings)
- └ Profile

7.2 Tab Switching

- `lazy: false` on MainNavigator — all tabs are pre-mounted
- Prevents wallet screen flicker on first visit
- Each tab remembers its own navigation state

7.3 Transition Animations

Transition	Animation
Auth stack pushes	slide_from_right
Onboarding/Welcome	fade
Main ↔ Auth	fade (via RootNavigator)
StoryViewer	fade + gesture disabled
Modals	slide (from bottom)
Side Drawer	custom translateX + opacity backdrop
GamePlayModal	fullScreen slide

8. Animation & Motion

8.1 Animation Inventory

Animation	Duration	Type	Purpose
Splash logo appear	600ms	Spring	Brand entry
Onboarding dot active	300ms	Timing	Position indicator
Story progress bar	5000ms per story	Timing (linear)	Story duration
Story pause	instant	setValue	Feedback
XP claim icon bounce	400ms	Spring	Reward feedback
XP claim float text	900ms up + 300ms fade	Timing	XP amount rise
XP card exit (claim)	420ms	Timing	Card removal
XP particle fly	700ms	Timing (parallel x+y)	Fly to destination
XP card bounce	2× Spring	Spring	Arrival feedback
Countdown number	900ms	Spring (1 → 1.4 → 1)	Countdown emphasis
Game result card	Spring	Spring (scale 0 → 1)	Win/loss reveal
Drawer slide in	280ms (delay)	Custom	Navigation transition

Drawer backdrop	250ms	Opacity	Focus overlay
Modal (bottom sheet)	Built-in	slide	Standard modal pattern

8.2 Animation Principles

Native Driver (`useNativeDriver: true`) All animations use the native driver where possible (transform, opacity). This offloads animation to the native thread for 60fps performance even during JS work.

Spring vs Timing

- `Animated.spring` : Physical feel, overshoots then settles. Used for: bounces, scale reveals, reaction feedback.
- `Animated.timing` : Precise duration. Used for: particles, fade-outs, linear progress.
- `Animated.sequence` : Chained animations. Used for: countdown 3 → 2 → 1, result reveal sequence.
- `Animated.parallel` : Simultaneous animations. Used for: particle X+Y travel, card fade+slide.

Animation Choreography — XP Claim Sequence

```
t=0ms    User taps Claim
t=0ms    Icon bounce begins (spring)
t=0ms    "+50 XP" float text rises (900ms)
t=0ms    Particle fires from claim button → XP card (700ms)
t=700ms  Particle arrives → XP card spring 1.4× → returns to 1.0
t=700ms  Particle fades (200ms)
t=1800ms Card fades + slides down (420ms)
t=2220ms Card unmounts (setGone(true))
```

8.3 Non-Animation Motion

- Pull-to-refresh:** Native RefreshControl with themed tint color
- Scroll snap:** SpotlightCarousel uses `snapToInterval` for satisfying swipe
- Keyboard avoidance:** CommentsScreen uses KeyboardAvoidingView for text input

9. Interaction Patterns

9.1 Touch Feedback

Component	<code>activeOpacity</code>	Effect
Post actions (like, save)	0.7	Subtle darken
Cards	0.8	Light darken
Tab bar items	0.85	Minimal
Primary CTA buttons	0.85	Darken
Story avatars	default	No feedback (story opens)

9.2 Optimistic Updates

Actions update the UI immediately without waiting for a server response:

- **Like/Unlike:** Count changes instantly; icon fills/unfills
- **Save/Unsave:** Bookmark fills/unfills; post appears/disappears in Bookmarks
- **Join/Leave Community:** Button state and member count update immediately
- **Follow/Unfollow:** Follower count updates + button text changes instantly

9.3 Form Validation

- Inline errors appear below each field
- Error borders highlight on invalid fields
- Submit buttons disabled until required fields are valid
- Real-time feedback (e.g. UPI ID regex as user types)

9.4 Empty States

Each screen has a styled empty state:

- Large emoji illustration
- Title ("No posts yet", "No bookmarks yet", etc.)
- Description copy
- CTA button where applicable

9.5 Loading States

- Button loading: spinner replaces label text
- Pull-to-refresh: native spinner at top of ScrollView
- OTP verify: 1.5s mock async to simulate network

9.6 Modal Patterns

Bottom Sheet (StreakModal, WalletModals)

- Full-screen modal with transparent background
- Backdrop (`rgba(0,0,0,0.55)`) behind sheet — tap to dismiss
- Sheet has rounded top corners (`radii.xl`)
- Handle indicator bar (38×4px) at top center
- Content scrollable inside sheet

Fullscreen Modal (StoryViewer, GameplayModal)

- `presentationStyle="fullScreen"` (StoryViewer: fade, Game: slide)
- No safe-area inset at bottom (edge-to-edge game board)
- Close button always accessible (top-left X or ← back)

PageSheet Modal (HistoryModal, LeaderboardModal in Games)

- `presentationStyle="pageSheet"` — covers ~90% from bottom
- Gesture-dismissible on iOS

9.7 Gesture Patterns

Gesture	Screen	Action
Pull down	Feed, lists	Refresh
Tap left half	StoryViewer	Previous story
Tap right half	StoryViewer	Next story

Press & hold	StoryViewer	Pause
Tap backdrop	Modals, Drawer	Close
Horizontal swipe	SpotlightCarousel	Advance card
Horizontal scroll	TrendChips, StoryRow, BadgeScroll	Browse

10. Theme System (Dark / Light)

10.1 Architecture

The theme is managed by `ThemeContext` using React's Context API:

```
ThemeContext = {
  isDark: boolean,
  colors: ColorPalette, // DARK_COLORS | LIGHT_COLORS
  toggleTheme: () => void
}
```

Every component calls `useThemeColors()` to get the current palette. Styles are computed with `makeStyles(colors)` inside a `useMemo` hook so they recompute whenever the theme changes.

10.2 Color Palette Type

```
type ColorPalette = {
  bg: { base; surface; card; elevated }
  primary; primaryLight; primaryDark
  cyan; cyanLight; cyanDark
  xpGold; xpOrange
  success; danger; warning; pink
  text: { primary; secondary; muted }
  border; borderHover
  glass; glassBorder
}
```

10.3 makeStyles Pattern

All screens and components use this pattern:

```
function makeStyles(c: ColorPalette) {
  return StyleSheet.create({
    container: { flex: 1, backgroundColor: c.bg.base },
    card: { backgroundColor: c.bg.card, borderColor: c.border },
    title: { color: c.text.primary },
    // ... all color-dependent styles use c.xxx
  });
}

function MyScreen() {
```

```
const colors = useThemeColors();
const styles = useMemo(() => makeStyles(colors), [colors]);
// ...
}
```

Styles that don't depend on colors (dimensions, flexbox, static sizes) are still defined in the same StyleSheet for a single source of truth.

10.4 Theme-Aware vs. Intentionally Fixed

Some elements intentionally remain dark regardless of theme:

Element	Color	Reason
Game play board (playModal)	#05050F	Immersive game UX
Community/event banner gradients	Dark hex values	Visual design intent
Story gradient backgrounds	Dark gradients	Full-screen immersive
StatusBar in game	style="light"	Dark background always

All other elements adapt to the current theme via `colors.xxx`.

10.5 StatusBar Handling

```
<StatusBar style={isDark ? 'light' : 'dark'} />
```

Applied on every screen that doesn't have a fixed dark background. Ensures system status bar icons are readable against both light and dark backgrounds.

11. Accessibility Considerations

11.1 Touch Targets

- Minimum tap target size: 40×40px enforced on all icon buttons (`iconBtn: { width: 40, height: 40 }`)
- CTA buttons: full-width or minimum 120px wide
- Bottom tab items: auto-sized by RN to be sufficiently tall

11.2 Color Contrast

- Dark theme: white-ish text (`#F1F0FF`) on `#070714` base — high contrast
- Light theme: dark text (`#1A1740`) on `#F8F7FF` — high contrast
- Brand purple (`#7C3AED`) used primarily for interactive elements, not for readability-critical text

11.3 Text Sizing

- Minimum font size: 10px (`fontSizes.xs`) used only for badges/meta
- Primary readable content: 13–15px
- No dynamic font scaling currently implemented

11.4 Keyboard Handling

- `KeyboardAvoidingView` on `CommentsScreen` for text input
- `Platform.OS === 'ios' ? 'padding' : 'height'` behavior
- Inputs trigger soft keyboard; forms scroll to keep focused input visible

11.5 Loading & Error States

- All async operations show loading feedback (spinner or `RefreshControl`)
- Error messages are inline and descriptive (not just red borders)
- OTP resend timer prevents confusion about expired codes

11.6 Visual-Only Indicators

- Liked state: color change (grey → pink) + count increment — not icon-only
- Saved state: outline → filled bookmark
- Story seen: grey ring vs gradient ring — relies on color; could add opacity distinction
- Streak days: text label + icon (✓ or 🔥) — not color-only

Appendix A — Screen Inventory

Screen	File	Tab / Stack
Splash	<code>auth/SplashScreen.tsx</code>	Auth
Onboarding	<code>auth/OnboardingScreen.tsx</code>	Auth
Welcome	<code>auth/WelcomeScreen.tsx</code>	Auth
Login	<code>auth/LoginScreen.tsx</code>	Auth
Register	<code>auth/RegisterScreen.tsx</code>	Auth
OTP	<code>auth/OTPScreen.tsx</code>	Auth
Forgot Password	<code>auth/ForgotPasswordScreen.tsx</code>	Auth
Home	<code>main/HomeScreen.tsx</code>	Home Stack
Notifications	<code>main/NotificationsScreen.tsx</code>	Home Stack
Comments	<code>main/CommentsScreen.tsx</code>	Home Stack
User Profile	<code>main/UserProfileScreen.tsx</code>	Home Stack
Story Viewer	<code>main/StoryViewerScreen.tsx</code>	Home Stack
Bookmarks	<code>main/BookmarksScreen.tsx</code>	Home Stack
Settings	<code>main/SettingsScreen.tsx</code>	Home Stack
Community List	<code>main/CommunityScreen.tsx</code>	Community Stack
Community Detail	<code>main/CommunityDetailScreen.tsx</code>	Community Stack
Events	<code>main/EventsScreen.tsx</code>	Direct Tab
Games	<code>main/GamesScreen.tsx</code>	Direct Tab

Wallet	main/WalletScreen.tsx	Direct Tab
Profile	main/ProfileScreen.tsx	Direct Tab

Appendix B — Component Inventory

Component	File	Description
Button	common/Button.tsx	Multi-variant CTA button
Input	common/Input.tsx	Text input with label + icon + error
CreatePostModal	common/CreatePostModal.tsx	Post composer modal
PostCard	home/PostCard.tsx	Social post display card
StoryRow	home/StoryRow.tsx	Horizontal story avatars
XPProgressBar	home/XPProgressBar.tsx	Level + XP progress
SpotlightCarousel	home/SpotlightCarousel.tsx	Featured content slider
SideDrawer	home/SideDrawer.tsx	Slide-in nav drawer
CustomTabBar	navigation/CustomTabBar.tsx	Bottom tab + FAB
StreakCard	gamification/StreakCard.tsx	Daily streak display

Appendix C — Context Providers

Context	Hook	Key State
ThemeContext	useTheme() / useThemeColors()	isDark, colors
AuthContext	useAuth()	isLoggedIn
PostsContext	usePosts()	posts[], actions
CommunityContext	useCommunities()	communities[], actions
WalletContext	useWallet()	balances, transactions[], settings
GamesContext	useGames()	matches[], stats